

BMAN24621 Business Data Analytics Coursework Project

Analysis of the World Happiness Scores by Country

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Analytical Plan

- I. **Business Understanding:** There was initial research made regarding the World Happiness Report, how the data was collected and the interpretation made in the report for guidance.
- II. **Data Preparation:** The data was cleaned and changed to the correct formats so it can be used in SAS Enterprise Guide, SAS Enterprise Mining Workshop, Excel. The data cleaning stage included changing column labels for SAS naming conventions, determining any missing values, and deciding what to do with them, determining the appropriate approach with missing data.
- III. **Data Understanding:** The data was analysed and with the help of the research it was interpreted to see what every column meant, how the calculations were made for the six factors. There was an analysis for missing variables, summary statistics, an exploratory analysis of happiness scores by region, the correlation between the happiness scores and the six factors and their interpretation, correlation of the happiness score with the positive and negative emotions in two countries, an analysis of the change in the factors affecting the happiness scores over the last decade for two countries.
- IV. **Modelling:** There were various models made with SAS Enterprise Guide, SAS Enterprise Mining Workshop and Excel. These included Generalized Linear Model, Regression, Decision Trees. Visualisation methods such as line graphs, bar graphs, box plots, pie charts were used to make the interpretation easier. This step formed connections between data to create information.
- V. **Evaluation:** In the evaluation step the graphs and the analysis made was analysed and evaluated both according to the new findings but also considering the previous finding and forming connections between information to create knowledge.
- VI. **Report:** The results and methods used were formed into a report, with a process easy to track.

Introduction

The data set that will be analysed in this report was acquired from Gallup World Poll and was used for the World Happiness Report. The World Happiness Report is a report evaluating how people would rate their happiness levels in more than 150 countries depending on various factors.

The datasets used in this report include the

- Happiness Scores by Country: includes happiness evaluation data for countries in 2022
- Happiness by Years: includes happiness evaluation data for countries for various years, the range of years changes depending on country
- Predicting Happiness 2022: this data set includes happiness evaluation data for various countries which will be used to predict their ladder scores

The objective of this report is to analyse happiness scores for each country according to 6 factors and do an explorative analysis and a cluster analysis regarding the ladder scores; use predictive models to estimate ladder scores of other countries.

The six factors that will be examined in the report are:

- GDP per Capita
- Social Support
- Healthy life Expectancy
- Freedom to make life choices
- Generosity
- Perceptions of Corruption

Primarily the data sets will be pre-processed in which they will be cleaned for any missing values or outliers; a preliminary analysis will be conducted to analyse the relationship between the six factors and the ladder score for the Happiness2022 data set; another analysis between the correlation with the positive or negative emotions for the data set HappinessbyYears.

Afterwards a Clustering Analysis will be done to cluster the countries into appropriate number of clusters in terms of the six happiness factors for Happiness 2022 data set and later the key characteristics for each country will be determined.

Finally, 2 predictive models will be built which will later be used as a baseline to predict happiness scores for another list of countries. Error measures will be used to evaluate the performance of the predictive models.

Exploratory Data Analysis

Data Pre-processing

Before getting into the analysis the data must be pre-processed for a better analysis and to increase the quality of the data. The data might need to be cleaned, missing values should be managed, and outliers should be removed for a more accurate analysis and to ensure data completeness.

Exploring the Datasets

There are three data sets that we will be working with:

- Happiness2022.xlsx
- HappinessbyYears.xlsx
- PredictingHappiness2022.xlsx

These data sets will be cleaned and pre-processed separately. There's currently no need for data integration as the data analysis will be made separately for each data set.

1. *Happiness 2022 Data Set*

The data set include 146 rows ranking countries according to their Ladder Scores. Data seems to be clean as there are numerical values other than the Country Names and the Regional Indicator and there are no missing values. The data is relevant and timeliness as the data is for the recent year 2022. The column titles have been changed for the SAS naming conventions.

2. *Happiness by Years Data Set*

The data set includes 2089 rows of data with missing values and the years for countries are not consistent as the starting year, ending year and the range differ. The years range from 2005-2021 so some data is not very up to date and the starting year and the ending year differs from country to country which raises questions regarding data completeness.

When the missing values in each column are investigated:

- Country Name: 0 rows,
 - Year: 0 rows,
 - Life Ladder: 0 rows,
 - Log GDP per Capita: 27 rows,
 - Social Support: 13 rows,
 - Healthy Life Expectancy at Birth: 58 rows,
 - Freedom to Make Life Choices: 32 rows,
 - Generosity: 80 rows,
 - Perceptions of Corruption: 113 rows,
 - Positive Affect: 24 rows,
 - Negative Affect: 16 rows,
- have missing values.

- It would harm the analysis to delete these rows as they still contain valuable information regarding other columns.
- The missing data will be left as *NULL*.

3. Predicting Happiness 2022 Data Set

This data set includes 12 rows of data and 9 columns, only the values in the Ladder Score column is missing as predictions will be made throughout this report regarding the values.

Exploratory Analysis

Happiness Scores by Regions

- The data set Happiness2022 was analysed for the distribution of ladder scores in different regions.
- A BoxPlot was made for easier visualisation of data regarding regional indicators.

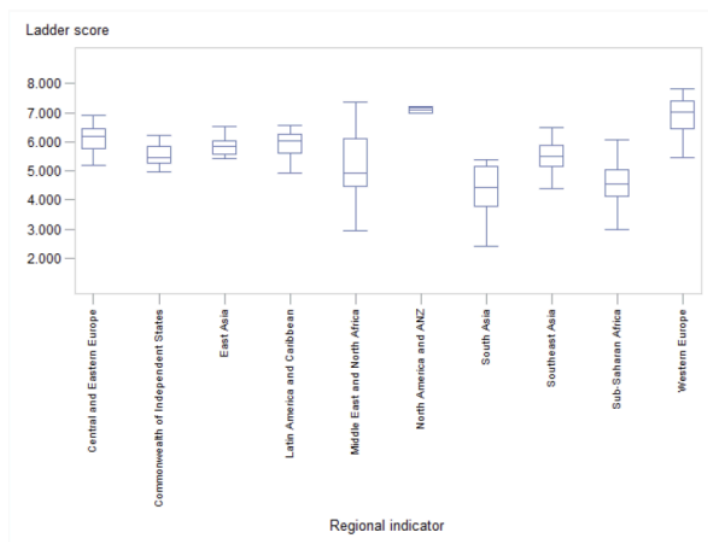


Figure 1: Boxplot Showing Ladder Scores by Regional Indicator

- The BoxPlot shows the 25 percentiles and there are 4 quartiles. The interquartile range (IQR) is the range from 25% to 75%.
- The data shows there's no outliers and the whiskers maximum and the minimum values.
- The Ladder Score distribution is wide in the Middle East and North Africa and North America and ANZ has the narrowest distribution of Ladder Scores.
- This will be further analysed by looking at the average ladder scores per region.
- A pivot table was created for each region showing the sum of ladder scores and the count of regional indicator, used to calculate the average ladder score per region.

Row Labels	Sum of Ladder score	Count of Regional Indicator	Average Ladder Score
Central and Eastern Europe	103.4348993	17	6.084405843
Commonwealth of Independent States	66.74289894	12	5.561908245
East Asia	35.2572999	6	5.87621665
Latin America and Caribbean	113.0566993	19	5.950352593
Middle East and North Africa	87.89150071	17	5.170088277
North America and ANZ	28.36380005	4	7.090950012
South Asia	25.59089994	6	4.265149991
Southeast Asia	48.88569975	9	5.431744417
Sub-Saharan Africa	156.7897007	35	4.479705736
Western Europe	144.8085999	21	6.895647617

Figure 2: Pivot Table Showing Region Indicators, Sum of Ladder Scores, Count of Regional Indicators

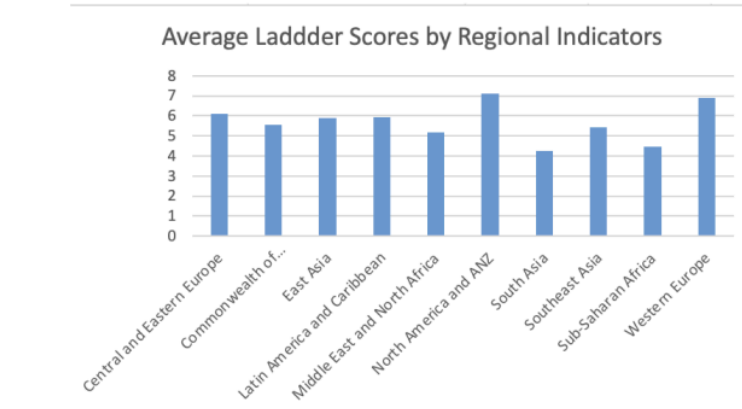


Figure 3: Average Ladder Scores by Regional Indicators Bar Graph

- A bar graph was used for easier comparison of ladder scores as the values are really close to each other.
- North America and ANZ has the highest average of ladder scores and South Asia has the lowest average of ladder scores by a noticeable difference.
- It can be seen on the BoxPlot that North America and ANZ has the shortest range of values with a relatively high average of 7.09. This indicates that the countries in this region would have high ladder scores.
- South Asia has a relatively wide range of ladder score values which indicates that there's both countries with higher and lower ladder scores and there's a big lower percentile with a few countries that have very low scores and the IQR is relatively lower than other regions.

Happiness Scores According to Factors

- The data set Happiness2022 was analysed in terms of how the six factors affect the ladder score. To do so, country name was set as the ID, and Ladder score was set as the target role for the purposes of the analysis.

Name	Role	Level	Report	Order	Drop	Lower Limit	Upper Limit
Country_name	ID	Nominal	No		No	.	.
Freedom_to_ma	Input	Interval	No		No	.	.
Generosity	Input	Interval	No		No	.	.
Healthy_life_ex	Input	Interval	No		No	.	.
Ladder_score	Target	Interval	No		No	.	.
Logged_GDP_pe	Input	Interval	No		No	.	.
Perceptions_of	Input	Interval	No		No	.	.
Regional_indica	Input	Nominal	No		No	.	.
Social_support	Input	Interval	No		No	.	.

Figure 4: Happiness2022.xlsx Data Roles

- “Stat Explore” was then used to generate summary and association statistics regarding the six factors affecting the ladder score which have been given the role inputs.

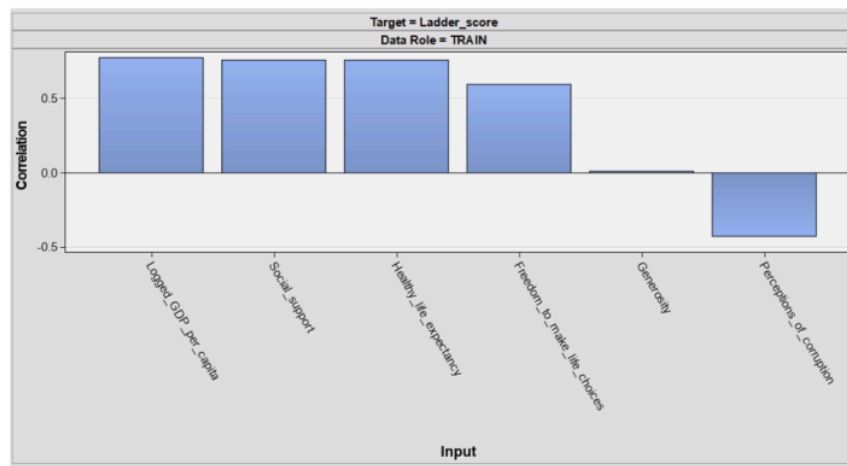


Figure 5: Correlation of the Six Factors to the Ladder Score Bar Graph

	Ladder score
Ladder Score	1
Logged GDP per Capita	0.776495319
Social Support	0.759034837
Healthy Life Expectancy	0.75862867
Freedom to Make Life Choices	0.592734947
Generosity	0.011218361
Perceptions of Corruption	-0.428251643

Figure 6: Correlation of the Six Factors to the Ladder Score Table

- Correlation analysis measures the strength of the linear relationships between variables and to what extent they affect each other.
- It can be seen that there's a strong positive correlation between the Logged GDP per Capita & Ladder Score; Social Support & Ladder Score; Healthy Life Expectancy & Ladder Score which indicates that an increase in these inputs would increase the ladder score.

- It's also examined that there's a negative correlation between the Perceptions of Corruption & Ladder Score. This indicates that an increase in perceptions of corruption would lead to a decrease in the Ladder Scores.
- The individual graphs for each factor vs. ladder score can be found in Appendix A.

Correlation of Happiness Scores with Positive and Negative Emotions in 2021

- Correlation of Ladder Score with the Positive and Negative Emotions was analysed using SAS Enterprise Guide.
- The Data was filtered for Ladder Score, Positive Emotions, Negative Emotions, and year was made equal to 2021.
- The following table shows the correlation between the values.

Pearson Correlation Coefficients, N = 119 Prob > r under H0: Rho=0		
	Negative_affect	Positive_affect
Life_Ladder	-0.62540	0.58818
Life_Ladder	<.0001	<.0001

Figure 7: Correlation Between Negative Affect and Positive Affect

- The findings show that there's a correlation between the happiness score and the positive and negative emotions even though it's not very strong.
- There's an inverse relationship between the negative emotions and the happiness score which indicates as the negative emotions increase the happiness score will decrease which is logical.

Analysing the Ladder Scores and Factors in the Last Decade for the UK and Finland

- Finland has been chosen for the comparison as it has the highest overall happiness score in 2022 and will be used for comparison with the UK.

1. Ladder Scores

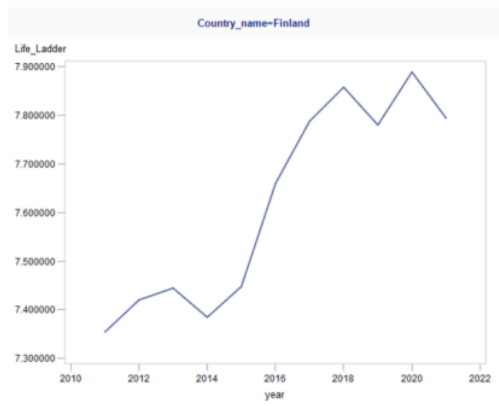


Figure 8: Finland Happiness Score vs. Year

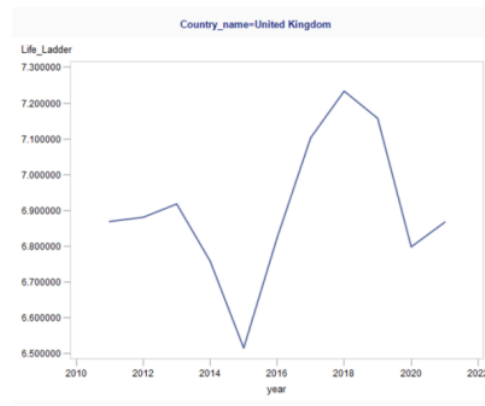


Figure 9: United Kingdom Happiness Score vs. Year

- Finland has had a recent increase in the happiness scores whilst the United Kingdom has had a recent decrease.
- The United Kingdom had a high happiness score in 2018.
- Finland's average happiness score is 7.61 while the UK's average happiness score is 6.91.

2. Logged GDP per Capita

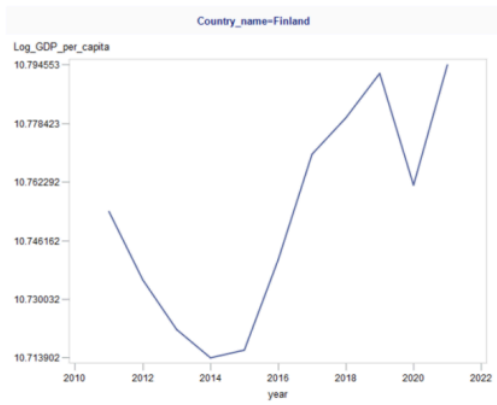


Figure 10: Finland Logged GDP per Capita vs. Year

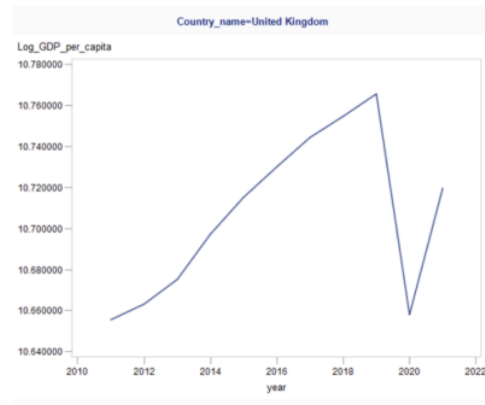


Figure 11: United Kingdom Logged GDP per Capita vs. Year

- There's been a decrease since 2018 in the United Kingdom and an increase in Finland since 2014.

3. Social Support

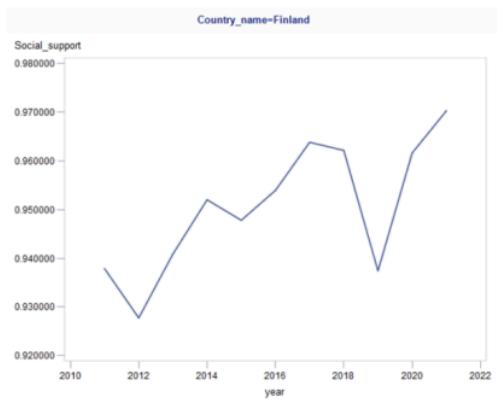


Figure 12: Finland Social Support vs. Year

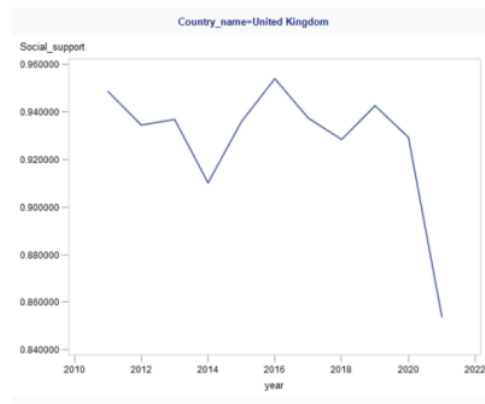


Figure 13: United Kingdom Social Support vs. Year

- There's been a noticeable decrease in the UK in 2021 and a noticeable increase in Finland since 2021.

4. Healthy life Expectancy

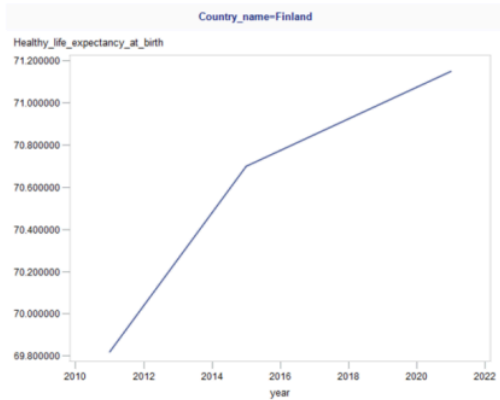


Figure 14: Finland Healthy Life Expectancy vs. Year

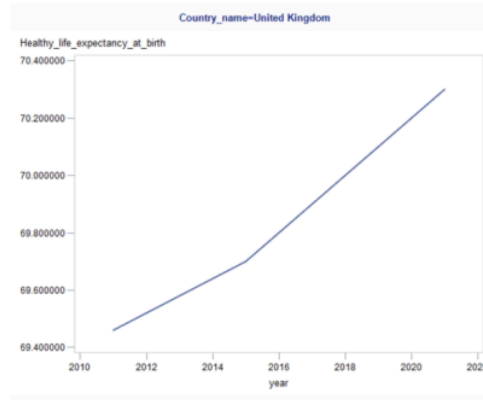


Figure 15: United Kingdom Healthy Life Expectancy vs. Year

- There's been a decrease in the slope of the line for Finland in 2015 whilst for the UK the slope of the line has increased suggesting a bigger increase in the life expectancy over the years.

5. Freedom to Make Life Choices

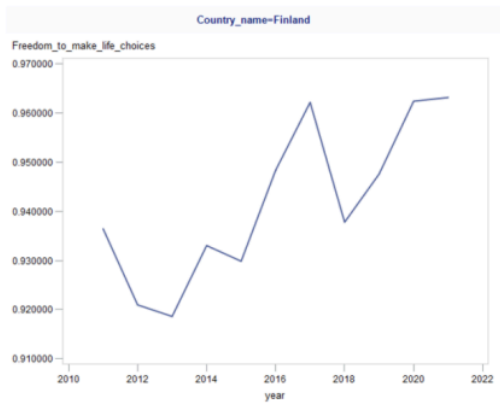


Figure 16: Finland Freedom to Make Life Choices vs. Year

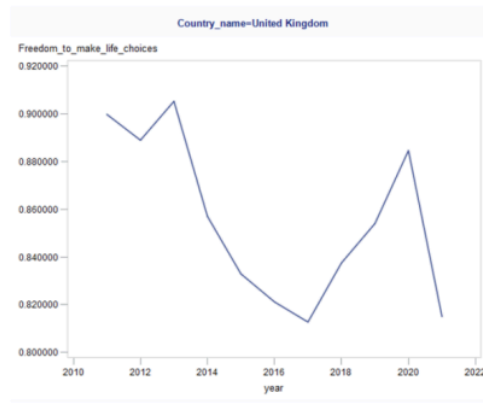


Figure 17: United Kingdom Freedom to Make Life Choices vs. Year

- Finland had a general increase in their freedom to make life choices whilst the UK had a decrease other than the period 2018-2020.

6. Generosity



Figure 18: Finland Generosity vs. Year

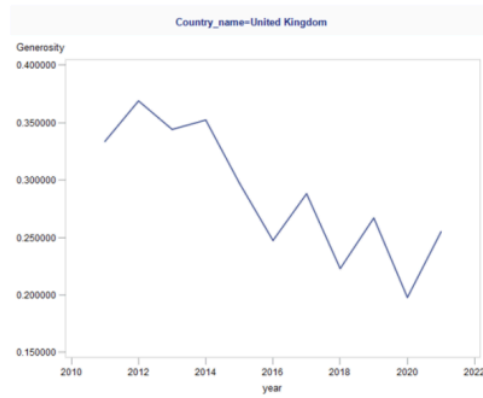


Figure 19: United Kingdom Generosity vs. Year

- The generosity scores for both companies are very fragile and there's a lot of ups and downs.
-

7. Perceptions of Corruptions

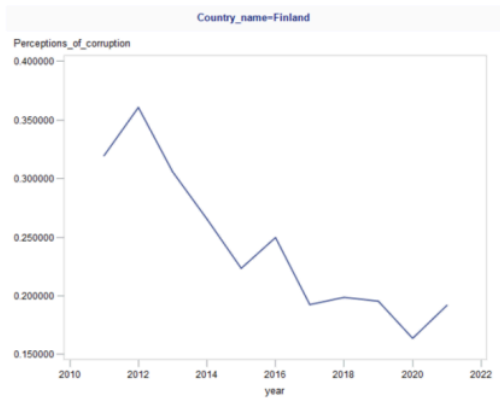


Figure 20: Finland Perceptions of Corruptions vs. Year

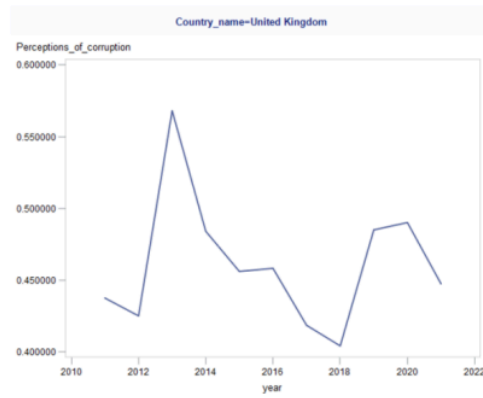


Figure 21: United Kingdom Perceptions of Corruptions vs. Year

- The perceptions of corruption have decreased in both countries since 2013 whilst in the UK it had increased in the period 2018-2020.

8. Positive Emotions



Figure 22: Finland Positive Emotions vs. Year

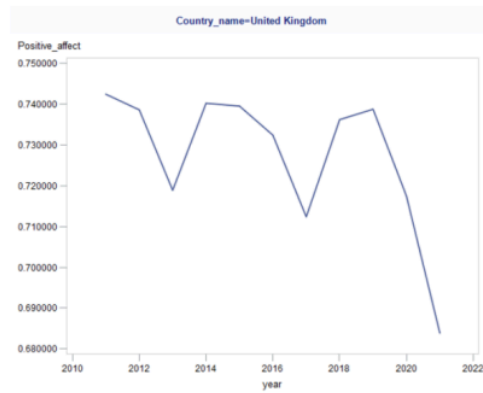


Figure 23: United Kingdom Positive Emotions vs. Year

- There's a noticeable decrease in the happiness score in the UK in 2021.

9. Negative Emotions

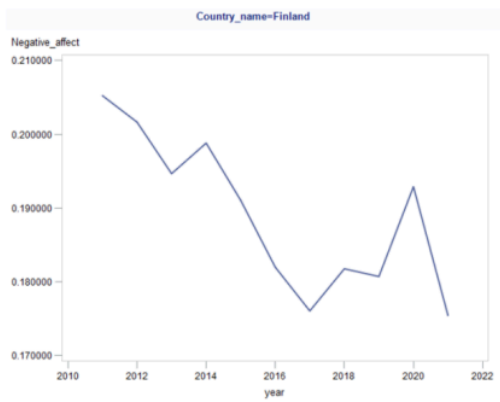


Figure 24: Finland Negative Emotions vs. Year

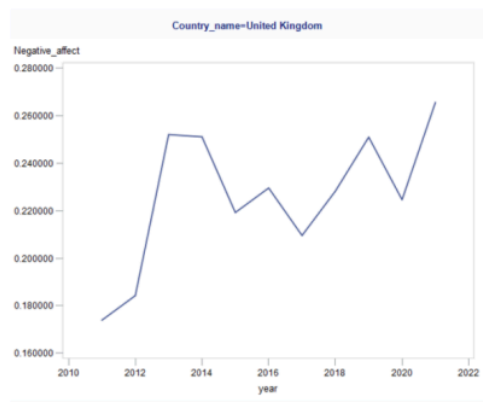


Figure 25: United Kingdom Negative Emotions vs. Year

- Whilst the graph for Finland shows that the negative emotions have generally decreased over the years, the UK shows that they have generally increased.

Cluster Analysis

Cluster Analysis will be applied to divide countries into clusters in terms of the six happiness factors.

- The maximum number of clusters was set to 5;
- Internal Standardization was set to Standardization.

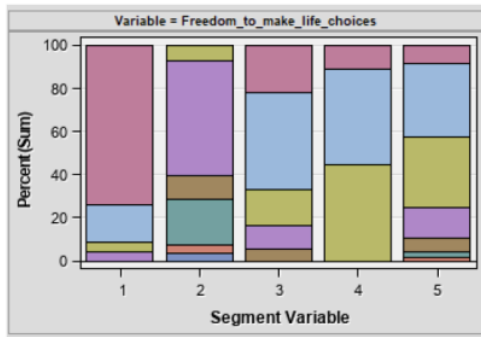


Figure 26: Segment Plot Freedom to Make Life Choices

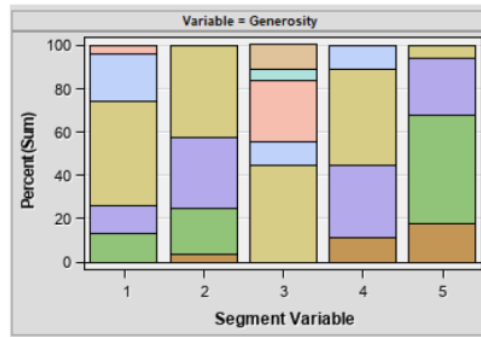


Figure 27: Segment Plot Generosity

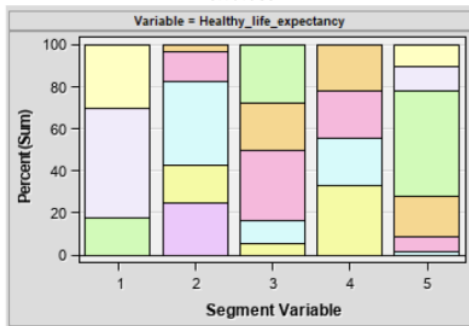


Figure 28: Segment Plot Healthy Life Expectancy

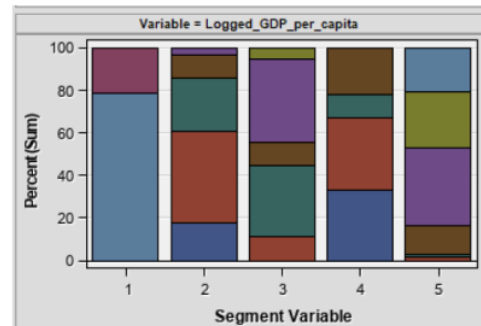


Figure 29: Segment Plot Logged GDP per Capita

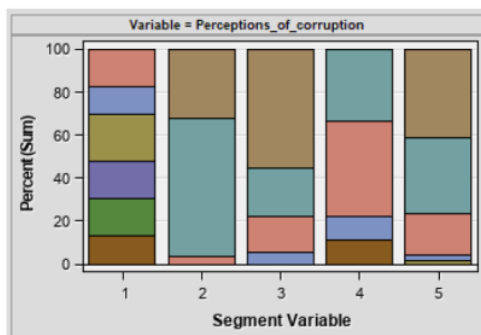


Figure 30: Segment Plot Perceptions of Corruption

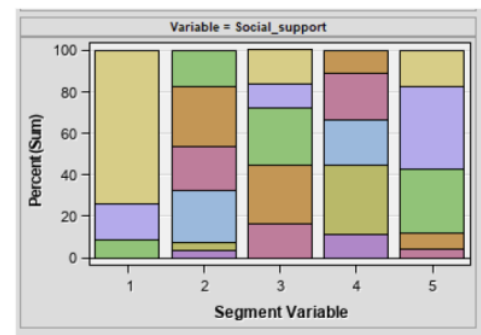


Figure 31: Segment Plot Social Support

- The distribution of values for each segment varies.

Freedom to make life choices	Generosity	Healthy life expectancy	Logged GDP per capita	Perceptions of corruption	Social support
0.902924	0.059741	72.55219	10.92906	0.420349	0.925854
0.657964	-0.01883	55.67974	8.09265	0.811516	0.707382
0.840239	0.225592	62.99836	8.968273	0.798953	0.809173
0.833189	0.015532	59.13177	7.884938	0.640819	0.613469
0.794223	-0.11371	67.80187	9.869194	0.789966	0.856653

Figure 32: Mean Statistics of Clusters 1, 2, 3, 4, 5

- Cluster 1 is the cluster with the highest happiness scores as it has the highest Logged GDP which has the highest correlation with the happiness score; highest social support which has the second highest correlation with the happiness score; highest life expectancy which has the third highest correlation with the happiness score.
- Cluster 1 also has the lowest perceptions of corruptions scores which supports the happiness score as they are inversely related.
- Cluster 4 has the lowest happiness score as the social support, logged GDP per capita are the lowest.

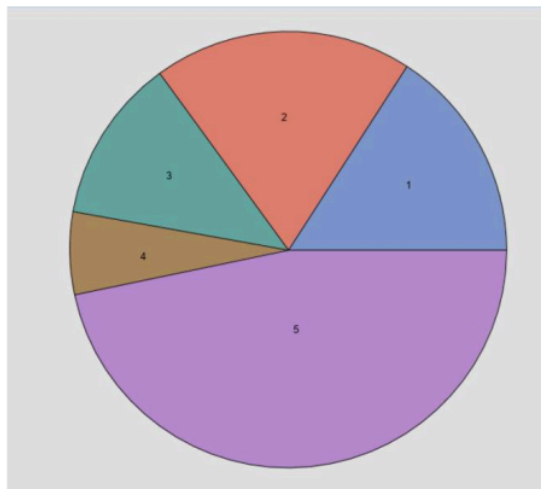


Figure 33: Segment Sizes for Each Cluster

Cluster Number	Frequency of Cluster
1	23
2	28
3	18
4	9
5	68

Table 1: Cluster Numbers and their Frequencies

- Cluster 4 which is assumed to have the lowest happiness score is also the one with the lowest frequency which indicates that this segment is the one with the lowest scores and is not as populated.
- The sorting for the clusters would be: 5>2>1>3>4

Predictive Modelling

- Three predictive models were used and evaluated by model comparison to choose the best model for making predictions regarding the Predicting Happiness 2022 data set.
- The data set Happiness 2022 was used for the model comparisons.
- The data set has been divided into training and validation data sets. Training being 80% and the validation set being 20%.

1. Decision Tree

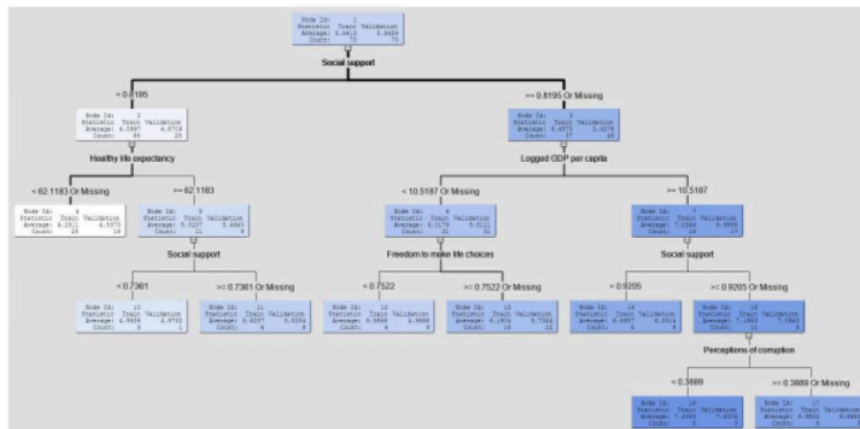


Figure 34: Decision Tree

- With the decision tree the data has been divided to hierarchies.
- ### 2. Generalized Linear Model
- Generalized linear models are another method for predictive analysis.
- ### 3. Regression
- Regression models investigate the relationship between an independent and a dependent variable.

Model Comparison

- The selection statistic was set to Mean Squared Error.
- This method was used for the comparison of all models that have been mentioned above.
- The red line shows the Target Mean while the blue line shows the predicted mean.

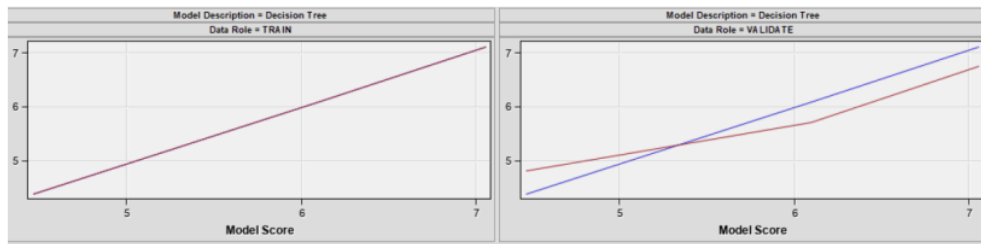


Figure 35: Score Distribution for the Decision Tree Model

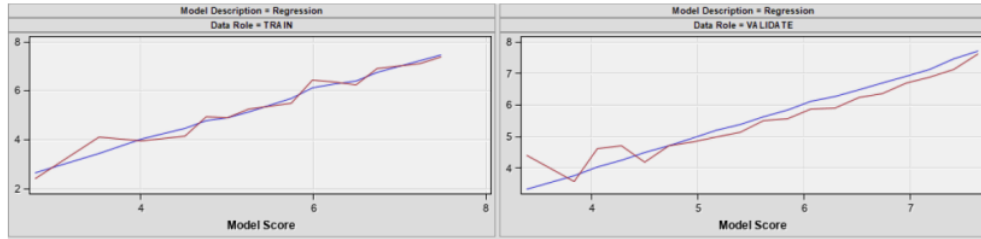


Figure 36: Score Distribution for the Regression Model

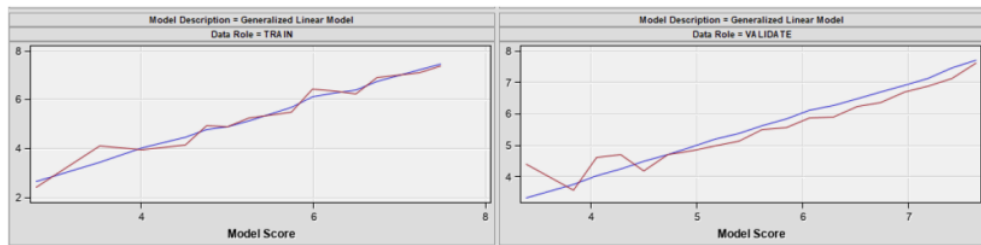


Figure 37: Score Distribution for the Generalized Linear Model

- Average Squared Error was used to evaluate the models. The ASE values for the models are stated on the table below.

Model	Train: Average Squared Error	Validate: Average Squared Error
Decision Tree	0.41135	0.6623
Regression	0.1498	0.4356
Generalized Linear Model	0.1498	0.4356

Table 2: Average Squared Errors According to the Predictive Models Table

- The average squared error of the regression model and the generalized linear model were equal whilst the average squared error for the decision tree was relatively higher.
- It can be suggested that the regression model or the generalized linear model are more reliable methods for prediction in this analysis as they have lower average squared error.
- As a result, regression was used by SAS Enterprise Miner as it was the most reliable one and the one with the lowest average squared error values.
- It can also be viewed visually through the training and the validation data sets the regression model provides the best fit.

The Predicted Values:

Country Name	Predicted Happiness Score
Uto	8.23
Wea	6.83
Ssaa	4.25
Seaa	5.37
Saa	4.99
Naa	7.08
Menaa	5.41
Laca	6.26
Eaa	5.43
Cisa	5.95
Ceea	6.29
Dysto	1.89

Table 3: Countries and their Predicted Happiness Scores

Data Analytics in Public Policy Making

The Covid-19 pandemic has had both negative and positive effects on people's wellbeing.

Positive and Negative Emotions During Covid

Looking back to the previous graphs regarding the happiness scores and the positive and negative emotions:



Figure 38: Finland Positive Emotions vs. Year

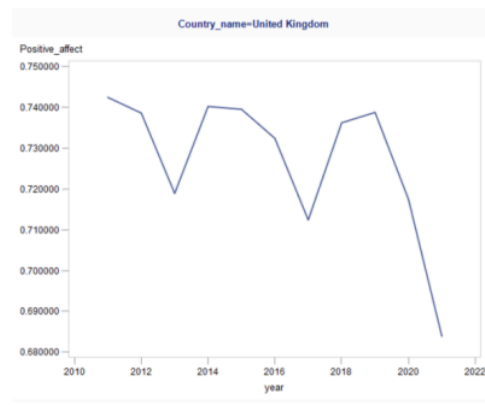


Figure 39: United Kingdom Positive Emotions vs. Year

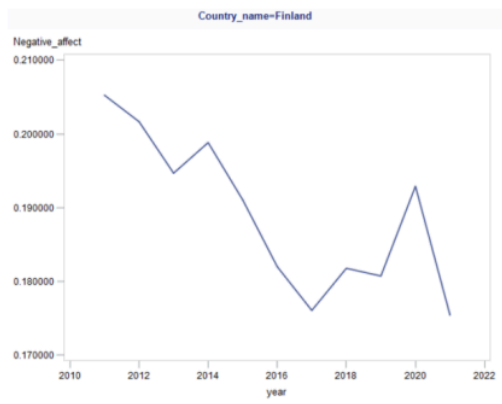


Figure 40: Finland Negative Emotions vs. Year

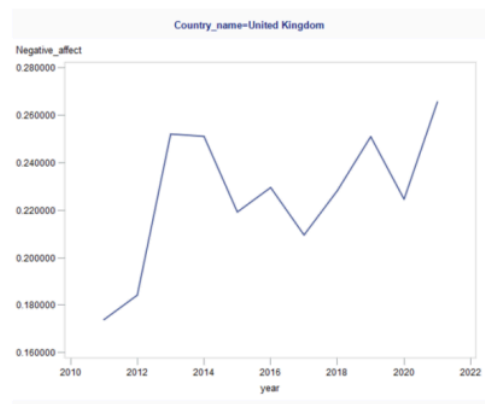


Figure 41: United Kingdom Negative Emotions vs. Year

- When the positive emotions are analysed, especially in the UK there's a fall of positive emotions during Covid-19 which might be related to the lockdowns and the spread of the virus.
- When the negative emotions are analysed in the UK there's a rise of negative emotions after the downfall from 2019 to 2020 which might indicate this could've been triggered by covid as well.
- The Negative emotions graph for Finland shows a strike at 2020 which could've been caused by the Covid-19 outbreak but the values drop in 2021.
- The difference of trends in these countries could've been triggered by the different policies governments have.

Government Policies

- Finland has multiple schemes during Covid to support people and businesses economically and there was also a lockdown due to the pandemic.
- The United Kingdom was very late to the lockdowns which caused thousands of people to die due to the coronavirus.
- The latened lockdown in the UK could've led to people losing their relatives, decreasing the happiness score.
- Finland's schemes to support people economically could've helped them keep the happiness scores high as people feel supported by their government.

Limitations

- There were various limitations regarding the data analysis one of the first being the missing data in the Happiness by Years data set.
- The Happiness by Years data set contained missing data and there are various ways to deal with missing data such as:
 - Replacing it with the average/mean
 - Replacing it with the predicted values
 - Leaving it null
 - Deleting the missing rows

- There's no perfect way to deal with missing data so in this report the missing data was left as it is.
- Replacing it with the mean or the predicted values would decrease the variability and if all rows containing missing data were deleted the analysis would be much narrower and more unreliable.
- Another limitation was that during training and validation data set arrangements there's no demonstration on why they were divided 80 to 20.
- The researcher is always biased, sometimes even unconsciously and therefore the research made regarding the policies could've been biased.
- As the data collected was from a sample population it might not represent the true population accurately.

Bibliography

Kang, H. (2013, May). The prevention and handling of the missing data. *Korean Journal of Anesthesiology*, 64(5), 402-406. Retrieved from <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3668100/>

Appendix

Appendix A: Individual Graphs for Factors vs. Ladder Score

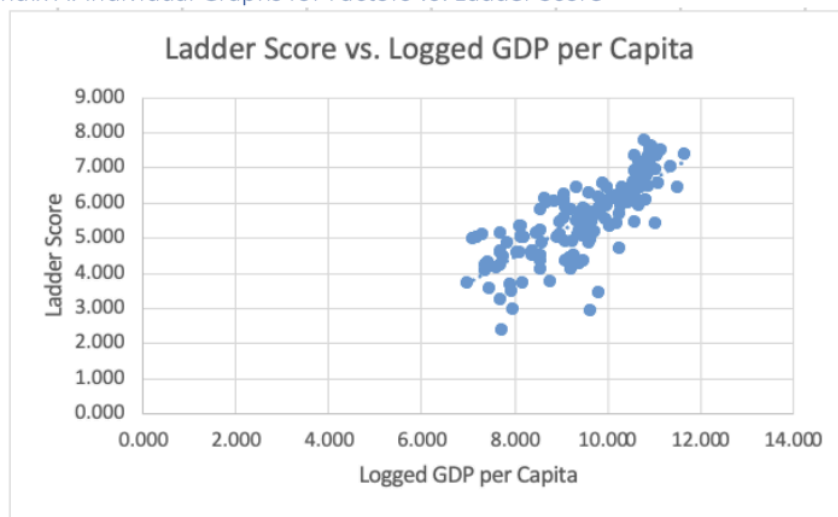


Figure : Ladder Score vs. Logged GDP per Capita

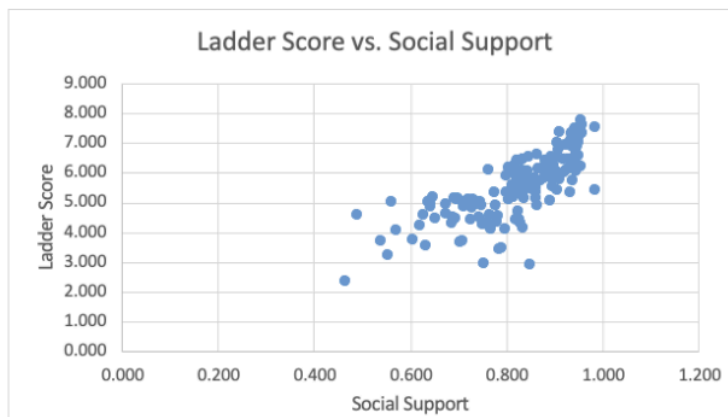


Figure : Ladder Score vs. Social Support

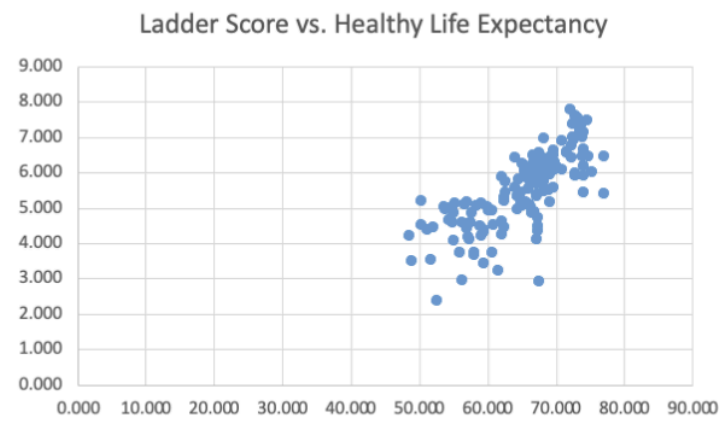


Figure : Ladder Score vs. Healthy life Expectancy

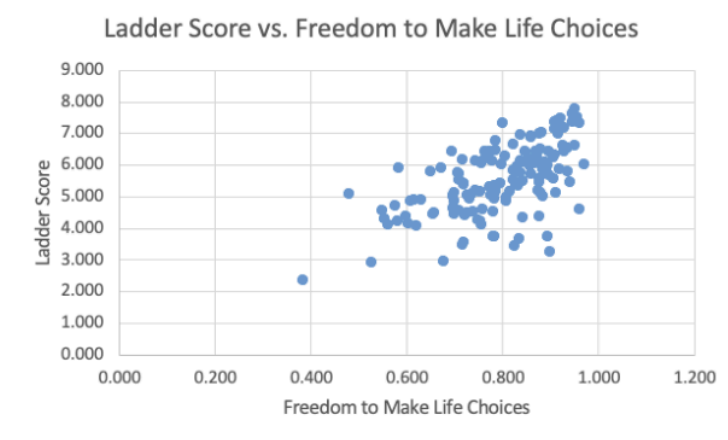


Figure : Ladder Score vs. Freedom to Make Life Choices

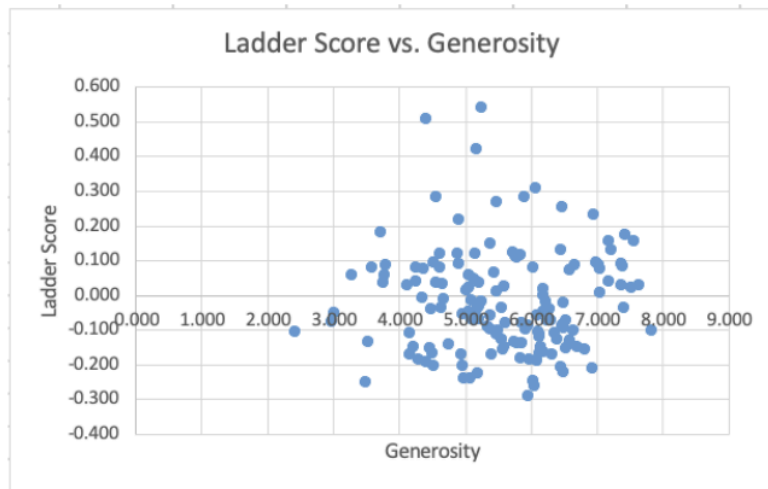


Figure : Ladder Score vs. Generosity

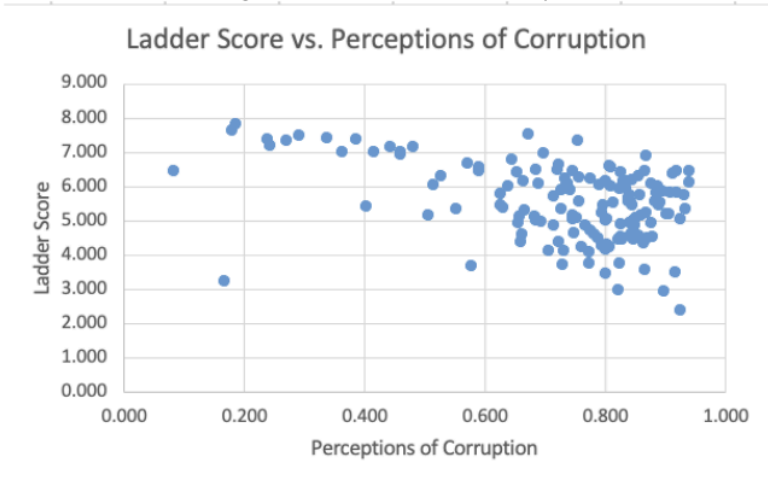


Figure : Ladder Score vs. Perceptions of Corruption